

Who Gets Protection? Geographic Inequality in Social Protection after Land Expropriation in China

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Abstract

Land in rural China is not only a productive asset but also a source of economic security. When farmland is expropriated, compensation determines not just how much wealth households recover but what replaces that security: a one-time cash payment, or continuing claims on pension, medical, and other social-protection systems. Using the rural samples of the China Household Income Project (CHIP) in 2013 and 2018, we study how this replacement protection was distributed across provinces and expropriation cohorts. Before the late 2000s it was strongly ordered by provincial development conditions: across the interquartile range of baseline fiscal pressure, households in weaker provinces were about 20 percentage points less likely to receive durable protection and 18 percentage points more likely to receive cash only. Among later cohorts these gaps are close to zero, and cross-province dispersion in protection breadth, durable protection, and cash-only compensation falls by about one-third. The decline survives adjustments for unequal province-cohort precision and household composition. We read this as a reorganization of realized protection's geography, not the effect of a single reform. Where access depends on place, social protection is itself a source of spatial inequality — one that weakened sharply as China's rural social protection expanded.

Keywords: land expropriation; social protection; geographic inequality; compensation; rural China; CHIP

JEL codes: H53; I38; O15; Q15; R58

1 Introduction

Estimates of how many rural Chinese have lost land to expropriation since the 1990s range from 50–60 million (Cai 2016) to more than 100 million (Zhan 2026). By 2006 the State Council had made social-security arrangements for those households a condition for approving an expropriation at all (State Council 2006). The reason these arrangements carry so much weight is that in rural China land is more than a productive asset. Contract and use rights generate agricultural income, but they also provide a fallback in old age and in other adverse states. Secure land-use rights can substitute for formal retirement protection in household decisions (Yu et al. 2010), while local “land for welfare” arrangements have explicitly linked land loss to pensions and other social benefits (Cai 2016). Land expropriation therefore removes both an income-producing asset and a source of economic security. How households are compensated matters not only for the amount transferred, but also for the asset they receive in return: a one-time cash payment, or a continuing claim on pension, medical insurance, minimum-income support, housing, or employment.

This distinction creates a distributional margin that is easy to miss when compensation is measured only in monetary terms. In a decentralized system, the institutions that replace lost security can depend on local fiscal resources, administrative capacity, and the reach of social-insurance programs. Households experiencing land expropriation may therefore receive fundamentally different forms of protection depending on where they live. One household may be connected to pension or medical insurance; another may receive cash and no continuing claim. Geographic inequality can thus arise before the conventional redistributive calculation begins: not only through differences in the amount of transfers received, but through unequal access to the institutions that provide protection in the first place. The characteristic that governs this access is, to a first approximation, location. The question in our title is therefore a geographic one: to ask who receives durable protection after land loss is to ask which provinces supplied it, and whether that set of provinces could be identified in advance from what those provinces looked like beforehand. This question is particularly relevant in China. Spatial differences have long been treated as constitutive of Chinese inequality rather than as a residual (Knight and Song 1993; Li and Sicular 2014; Luo et al. 2020), and the decentralization of social provision has produced substantial subnational variation in the design and implementation of welfare programs (Huang 2015). Land expropriation provides a setting in which these two features meet. Local governments historically had considerable discretion over how land-losing households were compensated, while central requirements concerning their long-run livelihood and social security strengthened during the 2000s. At the same time, the broader rural social-insurance system expanded rapidly. The resulting question is not simply whether compensation became more generous. It is whether the type of protection received after land loss remained systematically tied to provincial development conditions.

We study this question using the rural samples of the China Household Income Project

(CHIP) in 2013 and 2018. Both surveys identify households that experienced land expropriation and record the compensation associated with the household’s most recently reported episode. We harmonize these modules across waves and distinguish one-time cash compensation from continuing forms of protection. Our primary outcome, protection breadth, counts pension, medical, and Dibao protection, where Dibao is China’s means-tested minimum living guarantee (Song and Cheng 2024). We also use an indicator for any durable protection—including pension, medical insurance, Dibao, housing, or employment—and its economically natural counterpart, cash-only compensation.

One limitation bounds every result that follows, and it belongs here rather than in a robustness section. CHIP records whether a household received each form of protection, not the level of benefit that protection carries. What we measure is access to the institutions that provide security after land loss, not the security ultimately delivered, and the convergence we document is convergence in access. On the one dimension of generosity the survey does record—the compensation amount—the early geography compounds the pattern in form rather than offsetting it. Among households expropriated before 2008, moving across the interquartile range of baseline fiscal pressure is associated with roughly 0.37 log points less total real compensation. Weaker-baseline provinces supplied narrower protection and, in total, less money, so the architecture gradient is not a compensating differential in disguise.

The analysis asks two related questions, and they map onto the two standard notions of convergence, though not with equal exactness. First, was provincial inequality in realized protection systematically ordered by pre-existing development conditions, and did that ordering change across expropriation cohorts? This is the analogue of β -convergence: whether an initial provincial hierarchy stops predicting later outcomes. The parallel is not exact, because our conditioning variable is a fixed measure of baseline provincial development rather than the lagged level of the outcome itself, but the object it isolates is the same. Second, how large was the overall dispersion of protection across provinces, and did it fall? This is σ -convergence in the strict sense. The distinction matters. Provinces can remain far apart while ceasing to be ranked by any stable economic hierarchy, and in this setting that is close to what happens: both margins move, but the β -side moves much the further of the two.

Our main findings are as follows. First, the early geography of protection was strongly structured by baseline development conditions. Among households expropriated in 2000–03 and 2004–07, provinces with weaker baseline development conditions were substantially more reliant on cash-only compensation and substantially less likely to provide continuing protection. Moving from the 25th to the 75th percentile of baseline provincial fiscal pressure is associated with roughly a 20 percentage-point lower probability of receiving any durable protection and a 17–19 percentage-point higher probability of receiving cash only. The same broad ordering appears for protection breadth and when provincial GDP per capita is used to characterize the development hierarchy. The result is therefore better understood as a development gradient in realized

protection than as evidence for a narrowly identified fiscal mechanism. Second, that gradient largely disappears among later expropriation cohorts. For households expropriated from 2008 onward, the corresponding differences in durable protection and cash-only compensation are economically close to zero and change sign across cohorts. In a pooled specification that compares these cohorts with households expropriated before 2004, holding 2004–07 out as a separate transition period, the change in the baseline development gradient is large for protection breadth, any durable protection, and cash only; the transition period itself is indistinguishable from the earlier reference. In interquartile-range terms, later cohorts erase approximately 0.34 components of the early protection-breadth gradient, 19 percentage points of the durable-protection gradient, and 17 percentage points of the cash-only gradient. The change is therefore not merely a decline in the average distance between provinces. The particular provincial hierarchy that organized early compensation largely ceases to predict the protection received by later cohorts.

Third, overall geographic inequality declines as well. In a balanced panel of twelve provinces observed with usable support throughout the cohort comparison, mean absolute pairwise gaps are 36 to 38 percent lower for protection breadth, any durable protection, and cash-only compensation among the three cohorts beginning in 2008 than among the two earlier cohorts. Resampling households within province-cohort cells places the 95 percent interval for protection breadth at 19 to 47 percent (Appendix Table A15, Panel A), so the magnitude is best read as approximately one-third rather than as a precise figure. This compression is visible across alternative measures of compensation architecture and is not generated by the smaller size of early province-cohort cells. Artificially imposing the early sample-size structure on later cohorts still leaves a sizable decline in dispersion, while corrections that remove sampling variation imply an even larger decline in latent between-province heterogeneity. Reweighting later cohorts to match the observable composition of earlier households likewise preserves substantial geographic compression. Fourth, the timing points to a transition in the late 2000s but does not identify the causal effect of a single reform. Candidate boundaries in 2006, 2007, and 2008 produce the same qualitative weakening of the development gradient, while the contrast attenuates as the boundary is moved into 2009–11. Importantly, the gradient remains steep through 2006–07 even though national requirements concerning employment and social security for land-expropriated farmers had already tightened. The subsequent period combines further changes in land-expropriation governance with the expansion of rural pension and social-insurance infrastructure, the global financial crisis, and a large fiscal and credit response. We therefore use 2008 as a convenient empirical boundary between earlier and later cohorts, not as the treatment date of a discrete policy intervention.

These findings contribute to the literature in four related ways. First, research on land expropriation has shown that land loss affects income, saving, labor allocation, health, political attitudes, and subjective welfare, and that the form of compensation matters for these outcomes (Cai 2016; Qu et al. 2018; Wang et al. 2019; Zhao and Xie 2022; Huang et al. 2024; Zhan 2026).

We move one step earlier in the incidence of expropriation. Before asking how households respond to compensation, we ask how the architecture of that compensation is distributed. The institution replacing lost land security is itself an economically meaningful outcome. Second, the paper adds a distinct margin to the study of inequality and social protection in China. A long literature emphasizes spatial inequality, local implementation, and the redistributive effects of taxes and social-security transfers (Knight and Song 1993; Li and Sicular 2014; Huang 2015; Cai and Yue 2020; Guo and Zhao 2022; Yang and Zhao 2024). Our results show that inequality can arise at an earlier stage: in access to durable protection itself. Early in our sample, provincial development conditions strongly predicted whether land-losing households received continuing social protection or cash alone. Among later cohorts, that development hierarchy is largely absent. The relevant convergence is therefore not simply convergence in income or transfer amounts, but convergence in access to a form of economic security.

Third, the results distinguish formal institutional change from realized household incidence. National rules and provincial framework documents describe the obligations facing local governments, but program existence does not imply household receipt, particularly in a decentralized administrative system. The CHIP data allow us to observe the protection households report actually receiving after expropriation. This realized-incidence perspective reveals a sharper historical change than formal policy chronology alone: provincial heterogeneity remains, but its systematic ordering by the earlier development hierarchy largely disappears. Fourth, the paper contributes a measurement. Existing evidence on compensation architecture is either national but qualitative, or quantitative but confined to a small number of localities in developed regions. To the best of our knowledge this is the first nationally sampled quantitative description of realized compensation architecture across provinces and expropriation cohorts. Building it required harmonizing two survey waves whose land-expropriation modules differ in item numbering, in one response-option set, and in the presence of an item asked in only one of them; Appendix Tables A1 and A2 report that mapping in full. We then decompose the geographic dispersion of the resulting architecture into a component ordered by baseline provincial development and a residual, and show that the two behave very differently. The broader implication is that social protection is a distributional object before it becomes a redistributive instrument. Conventional analyses often ask how taxes, transfers, or insurance alter the distribution of resources conditional on participation. In decentralized systems, an additional source of inequality arises from who gains access to those institutions in the first place. Land expropriation makes this margin unusually visible because the loss of an asset with an insurance function creates an explicit need for replacement security. The Chinese experience shows that the geography of such access can change substantially even when local heterogeneity does not disappear.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting. Section 3 presents the CHIP data, compensation measures, and empirical approach. Section 4 documents the erosion of the provincial development gradient, and Section 5 examines the

compression of overall geographic dispersion. Section 6 reports measurement and design checks and summarizes the provincial policy archive. Section 7 concludes and discusses interpretation, policy implications, and the boundaries of the exercise.

2 Institutional Setting

2.1 The economic content of compensation

Land matters for this paper because it is both a productive asset and a source of self-insurance. Rural households hold contract and use rights rather than freehold title, but those rights support agricultural income and provide a fallback in old age and other adverse states—a margin that carries particular weight in a population where inequality among the elderly is itself unusually high (Hanewald et al. 2021). Consistent with this interpretation, Yu et al. (2010) show that secure land-use rights substitute for endowment insurance in rural household decisions, while Cai (2016) documents “land for welfare” arrangements that provide pensions and other benefits to land-losing villagers. Expropriation therefore removes both an income-producing asset and part of the household’s informal protection.

This makes compensation form economically meaningful. A one-time cash payment and a continuing social-insurance claim are not equivalent assets. Cash supplies liquidity. Pension-linked compensation creates a claim on future income, medical insurance reduces exposure to health expenditure risk, and employment, housing, or resettlement can provide other forms of durable support. The empirical object is therefore the architecture of replacement claims, not only the cash value of compensation. The two are not independent, however: if provinces that supplied little durable protection paid correspondingly more cash, the architecture gradient documented below would be a difference in the form of compensation rather than in its generosity. Because CHIP also records the amount received and the area taken, we test this directly in Section 6 and find no such offset. Zhan (2026) traces the institutional evolution of these forms across four decades; Qu et al. (2018) distinguish monetary from social-security modes. We study how these forms are distributed across provinces and expropriation cohorts.

2.2 From local experimentation to national social insurance

There was no single national reform assigning every expropriated household a common package. Localities experimented before the center imposed uniform obligations, and provinces and municipalities differed in eligibility, benefit formulas, financing, and whether affected households entered dedicated land-requisition schemes or broader pension programs.

Central requirements strengthened materially over 2004–07. In 2004, the State Council established a broad obligation to address the long-run livelihood and social-security needs of land-expropriated farmers (State Council 2004). The 2006 General Office guidance required local governments to establish appropriate employment and social-security arrangements and to

secure the necessary funding (General Office of the State Council 2006). Later that year, the State Council incorporated social-security costs into land-expropriation arrangements, allowed local governments to cover funding shortfalls from state-owned land-use revenues, and stated that land expropriation should not be approved when social-security funding was not in place (State Council 2006). A 2007 joint ministerial notice further strengthened social-security review and implementation requirements (Ministry of Labour and Social Security & Ministry of Land and Resources 2007). These measures made long-run livelihood protection an explicit part of the national framework while leaving substantial discretion over scheme design, eligibility, benefit formulas, and financing to local governments.

The broader social-insurance environment expanded after 2008. The New Rural Pension Scheme began as a national pilot in 2009 and explicitly required coordination with social-security arrangements for land-expropriated farmers (State Council 2009). The Social Insurance Law was adopted in 2010 and took effect in 2011, placing rural and urban social insurance within a broader national legal framework (Standing Committee of the National People’s Congress 2010). In 2014, the State Council unified the New Rural Pension Scheme and the Urban Resident Pension Scheme into the basic pension insurance system for urban and rural residents (State Council 2014). These were not dedicated land-expropriation reforms. Their relevance here is institutional: they broadened the social-insurance infrastructure with which local compensation arrangements could interact.

Two implications follow. First, province-wide policy dates are an incomplete description of provision, because pilots, financing choices, and administrative practice matter. Second, 2008 is not a clean treatment date: the shift we document sits inside a broad institutional and macroeconomic transition, which is what we characterize.

2.3 Related literature

Research on land expropriation uses microdata to study income, occupational change, consumption, saving, health, political attitudes, and other downstream outcomes after land loss (Wang et al. 2019; Zhao and Xie 2022; Huang et al. 2024). Our outcome is different. Wang et al. (2019) use CHIP 2013 to estimate the income and subjective-welfare consequences of expropriation and, in doing so, report the share of households receiving housing, medical, pension, and employment compensation—the same survey items from which we build our outcomes. There those components enter as regressors explaining household welfare. Here the bundle is itself the object to be explained. That is what we mean by moving one step earlier in the incidence of expropriation, and we do it on two harmonized waves rather than one.

A second group studies compensation form directly. Qu et al. (2018) distinguish monetary from social-security compensation in studying satisfaction, and Zhao and Xie (2022) link the terms of expropriation to local political trust. Two papers stand closest to ours.

Cai (2016) shows that land-losing villagers receive the welfare benefits that are least costly

to the local government, typically pension insurance, so that provision is selective in a way that tracks local fiscal capacity. Her evidence is case-based and subnational. Our contribution is not to disagree with her but to give that selectivity a magnitude and a date: across fifteen provinces it amounts to roughly twenty percentage points of durable protection over the interquartile range of baseline provincial fiscal pressure among households expropriated before 2008, and it has largely disappeared among later cohorts. What she documents as a mechanism, we quantify as a distribution and then observe eroding.

Zhan (2026) develops the concept of compensation regimes and dates a transition to a housing-centered regime in the late 2000s—the same period in which our provincial gradient erodes. The two exercises are independent: his periodization rests on policy texts and qualitative material, ours on household-reported receipt in a national sample, so the agreement between them is corroboration rather than duplication. They also measure different objects. He characterizes which compensation form was nationally dominant in each period; we measure how a given architecture was distributed across provinces and whether that distribution was systematically ordered. A shift in the national mix and a change in its geographic dispersion are separate facts, and the second does not follow from the first. Section 6 takes this up directly, since housing enters one of our measures and not the primary one.

A third literature studies the geography of Chinese welfare provision. Huang (2015) shows that social health insurance developed into distinct subnational systems shaped by local resources and central–local relations, and Chen et al. (2024) document convergence in a different institutional domain over a long horizon. Closest in design is Guo and Zhao (2022), who ask whether a province’s initial pension enrollment stops predicting its later enrollment and find convergence after 2012. Because that is structurally the question we ask, the differences matter. Their outcome is enrollment in a general pension program; ours is the bundle of protections a household reports receiving as a consequence of a specific event—what replaced an asset that had an insurance function. Their break is later than ours, so if both results stand, convergence in post-expropriation compensation preceded convergence in general pension coverage. Their measure of local capacity does not significantly predict enrollment, whereas baseline provincial development predicts compensation architecture strongly among our early cohorts. And they study whether the initial ordering persists; we separate that question from the behavior of dispersion itself, and find that the two margins move by different amounts.

Our measurement strategy sits within a body of work that uses CHIP to characterize distributional change and treats cross-wave harmonization as substantive rather than preliminary (Luo et al. 2020; Gustafsson and Wan 2020; Yang et al. 2021; Gustafsson and Sai 2023; Wang and Jv 2023; Yu et al. 2025). Li et al. (2020a) study the distributional role of public pensions in these data and Yang and Zhao (2024) document the growing redistributive weight of social security; both take the receipt of social protection as the distributional object, as we do. Li et al. (2020b) go further and treat data construction itself as the contribution. The dispersion

statistics we use also connect to work measuring spatial disparity in China directly (Lovely et al. 2019; Lai and Wang 2024).

Finally, comparing formal policy with realized receipt connects to work on implementation and take-up. Decentralized governance gives Chinese local authorities wide discretion in translating rules into provision (Xu 2011). Gallagher et al. (2015) document large enforcement differences in the Labor Contract Law; Huang and Kim (2020) show local adoption responding jointly to central pressure and local economic geography. Research on targeting and take-up establishes more broadly that program existence is not receipt (Alatas et al. 2012; Finkelstein and Notowidigdo 2019), and Han and Gao (2019) treat realized welfare receipt in rural CHIP data as the distributional outcome of interest. We measure that distinction in the land-expropriation setting.

3 Data, Measurement, and Empirical Approach

3.1 Sample

The household data are the rural samples of CHIP 2013 and CHIP 2018, repeated cross-sections designed to study income distribution in China. Both waves contain a land-expropriation module. For households reporting more than one expropriation, the questionnaire explicitly instructs respondents to answer for the latest episode. Throughout the paper, cohort therefore refers to the timing of the household’s *most recently reported expropriation*, not to a reconstructed first-event history.

The harmonized architecture sample contains **2,180 land-expropriated households in fifteen provinces**: 1,202 in 2013 and 978 in 2018, with reported expropriation years from 1995 to 2018. The cohort analysis begins in 2000 and uses **2,063 households** in five bins — 2000–03, 2004–07, 2008–10, 2011–13, and 2014 onward. The 2014+ cohort is observed only in 2018 by construction. The estimand pertains to households that remain in the CHIP rural sampling frame at the survey date.

The module also records which land was taken — arable land, the homestead, or land of other usage — and the sample retains all four affirmative categories. Arable land is involved in 93.6 percent of cases, 1.1 percent are homestead-only, and 5.3 percent are land of other usage. Because homestead expropriation is compensated through a partly distinct resettlement regime, Appendix Table A4 reports the composition and re-estimates the paper’s two headline objects on the 2,041 households whose expropriation involved arable land. Geographic compression remains between 29 and 36 percent and every estimated change in the development gradient is larger and more precisely estimated than in the full sample, so the results below are not an artifact of pooling distinct compensation regimes.

3.2 Realized protection

The compensation module records cash and several protection components associated with the most recently reported expropriation. Each of the six items from which our outcomes are built—cash, pension, medical, Dibao, housing, and employment—is worded as a consequence of the event: the questionnaire asks whether insurance was transacted, or a benefit provided, *due to this land expropriation*, rather than whether the household holds it (Appendix Table A1). What we measure is therefore event-attributed receipt and not general program coverage. Section 6 returns to what that distinction does and does not settle. We organize these variables around an explicit outcome hierarchy.

ProtectionBreadth is the primary outcome. It counts three durable protection components observed consistently across both waves: pension, medical protection, and Dibao. A value of zero indicates none of these protections; higher values indicate a broader durable package.

AnyDurableProtection is the binary headline measure. It equals one when the household reports at least one durable protection component in the harmonized architecture. **CashOnly** identifies packages that contain cash but none of pension, medical, or Dibao protection; no household so coded reports employment or housing protection either. These two measures provide the clearest economic contrast between a liquid payment and a package containing continuing protection.

We use two additional measures as supporting evidence. **SocialCore** indicates pension or medical protection. The *ImplementationIndex* combines SocialCore, cash plus social-core protection, pension, medical protection, protection breadth, and reversed CashOnly after standardization. Because several components overlap mechanically, the index is no longer a headline outcome; it is retained as a broad summary robustness measure.

Making the two waves comparable is a substantive step rather than a preliminary one. The 2013 and 2018 questionnaires are not identical: item numbering shifts, the K08.2 pension-type option set differs, and CHIP 2018 inserts an unemployment-insurance question with no 2013 counterpart, which we exclude from every outcome so that the architecture is comparable across waves. Neither wave’s data file carries variable labels, so the questionnaires are the authority on wording. Comparable care over cross-wave consistency is what makes repeated CHIP cross-sections usable for distributional work (Luo et al. 2020; Li et al. 2020b). Appendix Table A1 reproduces the exact question wording alongside the harmonized variable each item feeds, Appendix Table A2 gives both waves’ response options and the coding rule applied to each item, and Appendix Table A3 states every outcome as a formula over those items. Table 1 reports cohort means. National average receipt is not monotone over time; the paper’s object is the cross-province distribution rather than a claim of uninterrupted national expansion.

Table 1: Sample and compensation architecture by cohort

Cohort	N	Prov.	Cash only	Any durable	Pension	Social core	Breadth
2000-03	185	15	0.573	0.373	0.249	0.297	0.595
2004-07	330	15	0.509	0.433	0.203	0.376	0.582
2008-10	432	15	0.535	0.444	0.273	0.370	0.620
2011-13	572	15	0.407	0.568	0.414	0.498	0.899
2014+	544	15	0.461	0.478	0.342	0.382	0.702
All 2000+	2063	15	0.479	0.479	0.317	0.403	0.711

Notes. Land-expropriated rural households, CHIP 2013 and 2018, expropriation years 2000 and later. Entries are unweighted means. Protection breadth is a count of durable protection components; the remaining reported components are indicators. Cohort refers to the household’s most recently reported expropriation.

3.3 Geographic inequality

The geographic unit is the province. We residualize each outcome on household-head age and its square, sex, education, and survey wave, re-center the residualized outcome at the full-sample mean, and then average within each province-cohort cell. Cells with fewer than five households are excluded.

Changing province coverage across cohorts can mechanically alter dispersion. The head-line analysis therefore uses a **balanced twelve-province panel**: Anhui, Chongqing, Gansu, Guangdong, Henan, Hubei, Hunan, Jiangsu, Shandong, Shanxi, Sichuan, and Yunnan each have at least five architecture-valid households in all five cohorts. The balanced sample contains 1,901 households in usable cells. Beijing, Inner Mongolia, and Liaoning enter the broader sample only in later cohorts.

The public CHIP files used for this analysis do not contain a survey-weight variable that can be applied consistently to these rural expropriation modules. Household means are therefore unweighted; provinces are then weighted equally in the geographic statistics.

The primary dispersion statistic is the mean absolute pairwise gap among provincial adjusted means,

$$MAPG_c = \frac{2}{P(P-1)} \sum_{p < q} |\bar{Y}_{pc} - \bar{Y}_{qc}|, \quad (1)$$

which is the expected absolute difference between two provinces drawn from the observed set. We also report the cross-province standard deviation. Provinces receive equal weight because the estimand is geographic inequality across provincial units, not household-weighted national inequality.

A central measurement issue is that early province-cohort cells are smaller. We therefore report three additional objects. First, a matched- N stress test repeatedly reduces later cells to the province-specific size structure of the 2000–03 cohort. This deliberately injects early-period sampling noise into later cohorts and provides a conservative stress test, not a de-noised estimator. Second, a moment correction subtracts average within-cell sampling variance from the observed cross-province variance. Third, hierarchical random-intercept models estimate latent

between-province heterogeneity directly. Together these exercises distinguish genuine geographic dispersion from noise in estimated province means.

3.4 Baseline provincial development conditions

To characterize the structure behind geographic differences, we relate protection to a fixed provincial fiscal-pressure measure averaged over 2000–02,

$$FiscalPressure_p = 1 - \frac{LocalTaxRevenue_p}{LocalGeneralPublicBudgetExpenditure_p}. \quad (2)$$

Higher values indicate a weaker own-source fiscal position. We treat this variable as a marker of the broader provincial development hierarchy rather than as a separately identified fiscal mechanism: across the CHIP provinces it is strongly correlated with GDP per capita and with pre-crisis trade exposure.

The timing distinction is important. The 2000–02 measure overlaps the earliest 2000–03 expropriation cohort and is therefore not described as strictly predetermined for that cohort. For the 2004–07 cohort and every subsequent cohort, however, the baseline measure clearly predates the reported expropriation. The early development gradient does not depend on the overlapping cohort: it is at least as large and more precisely estimated in 2004–07. We therefore use 2000–03 as an earlier descriptive cohort and 2004–07 as the clean early benchmark for statements requiring temporal precedence.

For each cohort we estimate

$$Y_{ipt} = \alpha_c + \beta_c D_p + \lambda_t + \delta_w + \varepsilon_{ipt}, \quad (3)$$

where D_p is the fixed baseline measure, λ_t are expropriation-year effects, and δ_w is a survey-wave effect when the cohort spans both waves. Standard errors are clustered by province.

We also estimate a pooled slope-change specification on all 2,180 households,

$$Y_{ipt} = \alpha_p + \lambda_t + \delta_w + \theta_T (D_p \times Transition_t) + \theta (D_p \times Post_t) + \varepsilon_{ipt}, \quad (4)$$

where $Transition_t$ marks expropriations in 2004–07 and $Post_t$ marks 2008 onward. Province effects α_p absorb the level of D_p . The omitted category collects households expropriated before 2004, so θ is the change in the development gradient between the post-2008 cohorts and that pre-2004 reference, and θ_T is the corresponding change for 2004–07. We report θ as the headline slope change and θ_T alongside it, because whether the 2004–07 cohort belongs with the earlier or the later regime is an empirical question rather than something the specification should impose. With only fifteen province clusters, restricted wild-cluster bootstrap inference is the primary small-cluster procedure. Province-label permutation tests are reported as a complementary robustness exercise rather than as design-based randomization inference, since provincial

development conditions were not randomly assigned.

3.5 Provincial institutional archive

The second dataset records formal provincial frameworks for social protection after land expropriation. All fifteen CHIP provinces have a framework date. We use this archive to characterize institutional chronology, not to construct a causal instrument. Adoption timing is itself geographically structured: all late adopters in the sample are inland, while the coastal provinces adopt early or in the middle of the distribution. The archive therefore does not provide independent variation with which to separate formal institutionalization from provincial development.

4 β -Convergence: The Erosion of the Development Gradient

Where a household lived once predicted, with considerable force, what it received when its land was taken. Provincial differences in compensation architecture were ordered along a pre-existing development hierarchy, and that ordering largely disappears among later cohorts.

4.1 The early hierarchy is economically large

Figure 1 plots cohort-specific gradients on the fixed 2000–02 fiscal-pressure measure. Protection breadth, any durable protection, cash only, social core, and the broader implementation index all show a steep early relationship and a much flatter later profile. The pattern is especially transparent in the binary architecture measures. Across the fifteen CHIP provinces, the fiscal-pressure interquartile range is 0.207. For households expropriated in 2000–03 and 2004–07, moving from the 25th to the 75th percentile is associated with roughly a 20 percentage-point lower probability of any durable protection and a 17–19 percentage-point higher probability of cash-only compensation. In the three later cohorts, the corresponding differences are small and change sign across cohorts. Appendix Table A7 reports the underlying cohort-specific estimates.

The timing of the baseline measure does not drive this comparison. For 2004–07, the 2000–02 provincial measure unambiguously predates expropriation, and the early gradient is at least as strong as in 2000–03. The paper therefore does not rely on treating the overlapping 2000–03 cohort as a clean predetermined comparison.

4.2 Pooled slope changes

Table 2 reports θ from equation (4): the change in the development gradient between cohorts expropriated from 2008 onward and those expropriated before 2004. The coefficient is 1.647 for protection breadth, 0.933 for any durable protection, and -0.822 for cash only. Restricted wild-cluster bootstrap p -values are 0.0115, 0.0210, and 0.0215. In interquartile-range terms, the later cohorts erase about 0.34 protection components of the early breadth gradient, 19.3 percentage

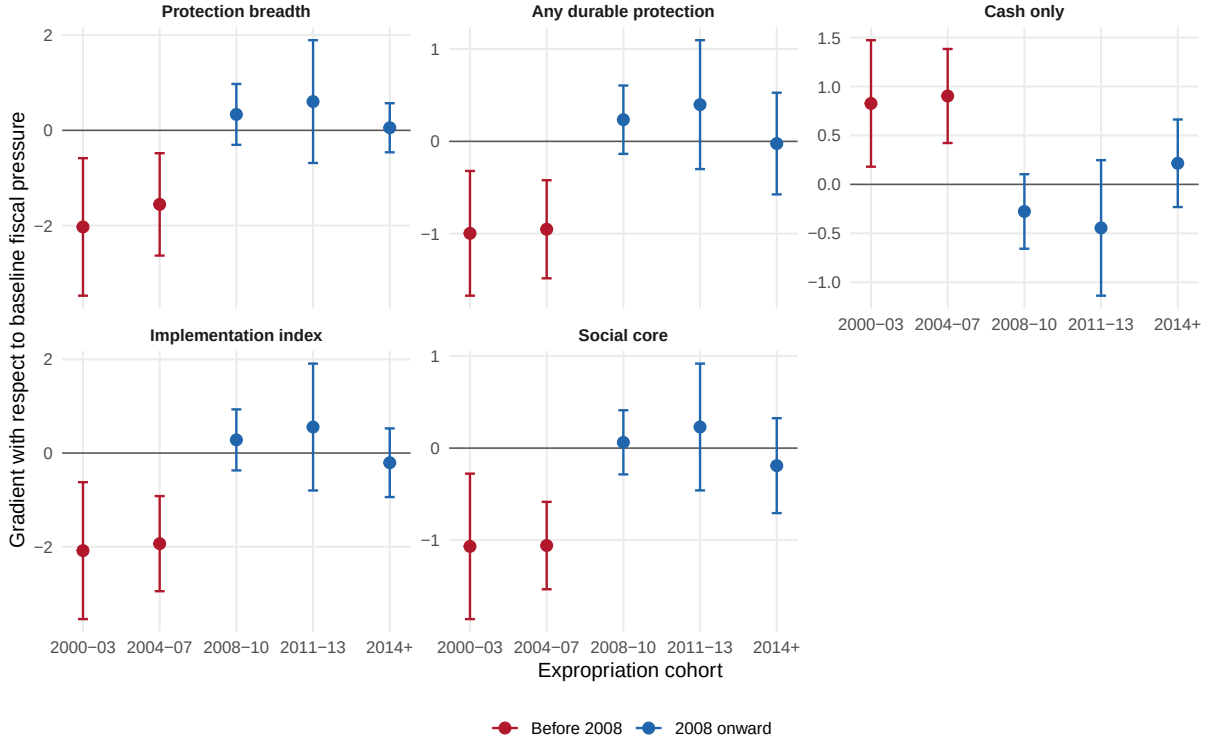


Figure 1: Development gradients in realized protection by expropriation cohort. Coefficients relate household outcomes to fixed 2000–02 provincial fiscal pressure within cohort, with expropriation-year and survey-wave effects. Higher fiscal pressure denotes a weaker baseline own-source position.

points of the any-durable gradient, and 17.0 percentage points of the cash-only gradient.

The 2004–07 cohort belongs with the earlier regime rather than the later one. The transition coefficient θ_T is -0.072 for protection breadth, -0.233 for any durable protection, and 0.261 for cash only, none distinguishable from zero at conventional levels. The gradient facing households expropriated in 2004–07 is therefore statistically indistinguishable from the gradient facing those expropriated before 2004, even though central requirements were being tightened over exactly that window. Pooling the two earlier cohorts when describing the result is an empirical finding here, not an assumption built into the specification.

Province-label permutation tests are more conservative. Their p -values are 0.0880 for protection breadth, 0.0600 for any durable protection, and 0.0585 for cash only. We report these alongside the wild-cluster results rather than replacing them: fiscal pressure is an observed provincial characteristic, not a randomized treatment, so the permutation exercise is a robustness check on the province-level pattern rather than design-based randomization inference. The implementation index shows a slope change of the same sign and significance, while social core alone is estimated less precisely. Protection breadth, any durable protection, and cash only therefore carry the main result, with the composite index as a summary measure.

Table 2: Change in the provincial development gradient

Outcome	$\hat{\theta}$	Cluster SE	Cluster p	Wild p	Permutation p	IQR magnitude
Protection breadth	1.647	0.702	0.019	0.0115	0.0880	+0.34 components
Any durable protection	0.933	0.357	0.009	0.0210	0.0600	+19.3 pp
Cash only	-0.822	0.278	0.003	0.0215	0.0585	-17.0 pp
Implementation index	1.473	0.703	0.036	0.0345	0.1430	+0.31 units
Social core	0.654	0.420	0.120	0.2475	0.2855	+13.6 pp

Notes. $\hat{\theta}$ is the coefficient on $D_p \times Post_t$ in equation (4), estimated on all 2,180 households with province, expropriation-year, and survey-wave effects. It is the change in the baseline fiscal-pressure gradient between cohorts expropriated from 2008 onward and the omitted reference of cohorts expropriated before 2004; the 2004–07 transition period enters through a separate interaction, reported in the text. Fifteen province clusters; restricted wild-cluster bootstrap and province-label permutation tests each based on 1,999 draws. Both resampling p -values lie on a 1/2,000 grid and are reported at four decimals, because several fall on a three-decimal half-way point where rounding conventions differ. “IQR magnitude” converts $\hat{\theta}$ into the change associated with moving from the 25th to the 75th percentile of baseline fiscal pressure across the fifteen provinces.

4.3 The change occurs in the late 2000s

Appendix Figure A3 and Appendix Table A8 report a candidate-breakpoint profile. Boundaries placed in 2006, 2007, or 2008 all yield a gradient change that clears conventional levels under restricted wild-cluster inference for every one of the five outcomes; by 2011 none does. The result is therefore not specific to a single threshold. The estimated change is largest at 2008 and declines on both sides of it, which is consistent with a transition concentrated in the late 2000s: a boundary set too early assigns 2006–07 households to the later group, one set too late assigns 2008 households to the earlier group, and either misassignment dilutes the contrast. We do not read the peak as independent confirmation of 2008. A profile maximum identifies the boundary that best separates the two regimes in these data, which is not the same as establishing that a discrete change occurred in that year.¹ The fine cohort split tells the same story: the implementation-index gradient is -2.083 , -1.885 , and -1.884 in 2000–03, 2004–05, and 2006–07 before moving to $+0.281$ in 2008–10 (Appendix Table A9). We therefore describe the transition as a sharp weakening around the late 2000s and use 2008 as a convenient empirical boundary. We do not interpret the breakpoint profile as evidence that a single reform took effect discretely in that year.

Figure 2 summarizes the geographic reordering in the primary outcome. Provinces are ranked by the fixed baseline fiscal-pressure measure. The early columns display a clear development-ordered pattern; later columns retain substantial local heterogeneity but much less systematic ordering.

¹The single-break coefficient at 2008 is larger than the pooled estimate in Table 2 — 2.046 against 1.473 for the implementation index — because the two specifications use different reference periods. The pooled specification holds 2004–07 out as a separate transition period and measures the later change relative to pre-2004 cohorts, whereas the single-break version pools both steep-gradient cohorts into the reference group.

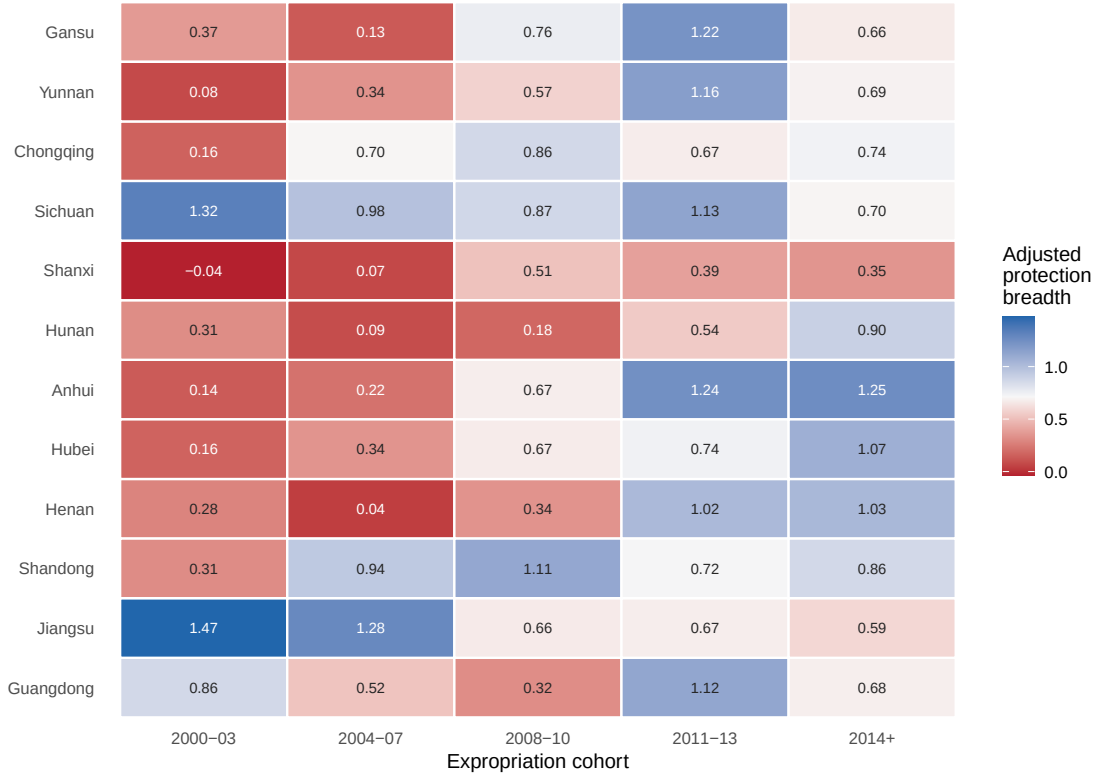


Figure 2: Realized protection by province and expropriation cohort. Demographically adjusted protection breadth on the balanced twelve-province panel, with provinces ordered by fixed 2000–02 fiscal pressure, weakest baseline position at the top. The ordering variable is fixed independently of the cohort outcomes; the fill is centered on the full-sample mean.

5 σ -Convergence: The Compression of Geographic Dispersion

The development hierarchy is one way provinces differed. It is not the only way. Provinces can stop being ordered by an economic ranking while remaining just as far apart, so the total spread of provincial outcomes is a separate question from the structure behind it. This section takes up the spread.

5.1 Large early disparities

Compensation architecture differs substantially across provinces among the early expropriation cohorts. Figure 3 shows the cohort path of mean absolute pairwise gaps on the balanced twelve-province panel. Protection breadth has MAPG values of 0.522 and 0.478 in 2000–03 and 2004–07, compared with 0.310, 0.343, and 0.280 in the three later cohorts. Any durable protection falls from 0.283 and 0.314 to 0.179, 0.189, and 0.202, and cash-only dispersion from 0.295 and 0.265 to 0.189, 0.184, and 0.160. The broader summary measures show the same ordering. For all five outcomes, every later cohort lies below both early cohorts. Appendix Table A6 reports the full cohort paths.

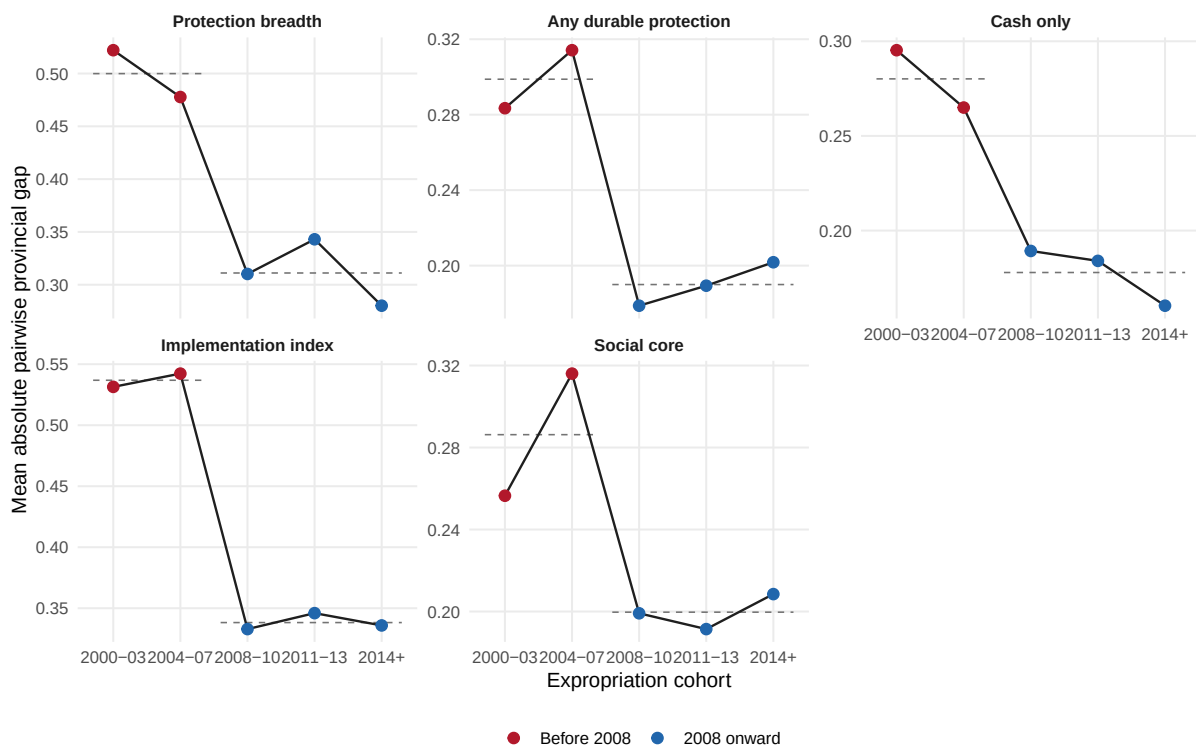


Figure 3: Geographic dispersion in realized protection across expropriation cohorts, balanced twelve-province panel. Points are mean absolute pairwise gaps among adjusted provincial means; dashed lines show the averages for the two earlier and three later cohorts.

The result is not confined to the composite measures. Averaging across the two early and three later cohorts, MAPG falls by 37.8 percent for protection breadth, 36.4 percent for any durable protection, and 36.5 percent for cash-only compensation. The decline is therefore not specific to the composite implementation index or to fine distinctions among individual program components.

5.2 The decline survives precision correction

The early cells are smaller, so a direct comparison of estimated province means could in principle overstate early heterogeneity. It does not. Table 3 compares five measures of compression. The raw MAPG decline is 30–38 percent across the five outcomes. When later province-cohort cells are repeatedly reduced to the early sample-size structure, the estimated compression remains 18–24 percent. Because this exercise adds sampling noise to the later period, it is best read as a conservative stress test.

Corrections that remove sampling variation move in the opposite direction. The moment-based de-noised standard deviation implies 34–46 percent compression, while hierarchical estimates of latent province heterogeneity imply 40–47 percent compression. For protection breadth

Table 3: Geographic compression under alternative precision treatments, balanced twelve-province panel

Outcome	Raw MAPG	Raw SD	De-noised SD	Hierarchical τ	Matched- N stress test
Protection breadth	37.8	41.3	46.4	45.8	23.4
Any durable protection	36.4	35.9	37.8	44.5	22.9
Cash only	36.5	34.5	35.7	43.9	22.2
Implementation index	37.0	40.7	45.5	46.6	23.6
Social core	30.2	32.1	34.1	39.8	18.0

Notes. Entries are percentage declines from the average of the two earlier cohorts to the average of the three later cohorts. Raw MAPG and raw SD use adjusted provincial means directly. De-noised SD subtracts estimated within-cell sampling variance before computing between-province dispersion. Hierarchical τ is the estimated latent between-province standard deviation from random-intercept models. The matched- N column repeatedly reduces later province-cohort cells to the early cell-size structure; it is a stress test that injects early-period sampling noise into later cohorts, not a corrected point estimate.

the corresponding estimates are 37.8 percent in raw MAPG, 23.4 percent in the matched- N stress test, 46.4 percent after de-noising, and 45.8 percent in the hierarchical model. Any durable protection and cash only display the same pattern. Sampling error therefore does not generate the observed compression; if anything, it masks part of the decline in latent geographic heterogeneity.

Inference depends on the population to which the result is generalized. When the twelve provinces are treated as the fixed geographic set of interest and households are resampled within province-cohort cells, confidence intervals for compression exclude zero for all five outcomes. When the twelve provinces themselves are resampled as if they were draws from a broader population of provinces, intervals widen substantially: cash only and any durable protection still exclude zero, while the interval for protection breadth includes it (Appendix Table A15, Panel A). With twelve clusters this exercise has little power, so we read it as a limit on how far the estimate generalizes rather than as evidence against the within-sample result. We state the main geographic result for the balanced set of CHIP provinces and do not treat the twelve-province figure as a precise estimate for all Chinese provinces.

The result is also not driven by a single province. Leave-one-province-out estimates preserve the sign of compression for every outcome (Appendix Table A15, Panel B). The broader available province set yields smaller compression because Beijing, Inner Mongolia, and Liaoning enter only in later cohorts and add post-period dispersion, but the direction remains the same (Appendix Table A6).

5.3 Total dispersion falls; systematic sorting nearly vanishes

The two results are economically distinct, and the collapse in sorting is the larger of them. Provinces remain heterogeneous after 2008, but baseline development explains very little of that remaining variation. On the balanced panel, baseline fiscal pressure explains 23.7 and 30.9 percent of cross-province variation in protection breadth in the two early cohorts, compared with 0.1, 2.9, and 0.3 percent in the three later cohorts. Baseline GDP per capita traces the same profile, with R^2 values of 0.220 and 0.359 early and 0.000, 0.073, and 0.034 later. The

implementation index behaves the same way, with 21.5 and 37.9 percent explained early against 0.2, 1.1, and 0.1 percent later. Appendix Figure A1 plots these shares for every outcome and baseline measure. The decline in systematic ordering is therefore much larger than the decline in total geographic dispersion.

This pattern is not specific to one provincial proxy. GDP per capita reproduces the broad early-to-later change, while pre-crisis trade openness is highly correlated with the fiscal-pressure measure. The data therefore identify a development-structured geography of protection more clearly than they identify a single fiscal or trade mechanism behind that geography.

6 Robustness, Measurement, and Formal Chronology

The main findings survive the measurement checks that are feasible in CHIP. The remaining limitations define the population and event history to which the estimates apply; they do not overturn the geographic pattern.

6.1 Retrospective lag

Recall horizons are longer for early expropriations. A common short-lag restriction cannot retain the 2000–03 cohort, whose minimum possible lag is already ten years, so the clean comparison uses 2004–07 as the early window. Compression does not shrink when long-recall observations are removed. For the implementation index it rises from 37.6 percent in the unrestricted common-window comparison to 42.0 percent within ten years and 44.7 percent within eight years (Appendix Table A10; Appendix Figure A2 shows the lag distribution). The early development gradient also remains large under the same restrictions (Appendix Table A10, Panel B). The result is therefore not concentrated in the longest retrospective lags.

6.2 Composition

Household heads in later cohorts are older at expropriation. Reweighting later cohorts to the early distribution of sex, education, age at expropriation, and its square eliminates the main observable imbalance without extreme weights. Geographic compression remains positive, in the 26–33 percent range in the original architecture measures (Appendix Table A11). Composition changes affect the exact magnitude but do not account for the direction of the result.

6.3 Coarse versus fine measurement

The main pattern survives when compensation is reduced to the broadest economically meaningful distinction. MAPG falls by 36.5 percent for cash only and 36.4 percent for any durable protection (Appendix Table A12). These coarse outcomes do not require respondents to recall the full package composition, and they reproduce the same early development ordering and

later flattening. The result is therefore not an artifact of the implementation index or of fine component coding.

6.4 Compensation value

Treating a continuing protection claim and a one-time cash payment as different assets raises an obvious question: did provinces that supplied less durable protection compensate with larger payments? They did not. The module records the amount the household received, which we deflate to 2018 yuan with event-year CPI, and the area taken. Between 85 and 94 percent of households in every cohort report an amount, so conditioning on a positive figure does not select strongly. Among households expropriated before 2008, moving from the 25th to the 75th percentile of baseline fiscal pressure is associated with 0.37 log points *less* total real compensation ($p < 0.001$), and with 0.11 log points less per mu of land taken, although the latter is not distinguishable from zero. Among later cohorts both gradients turn positive and imprecise. The early cash-only geography was therefore not a compensating differential: weaker-baseline provinces gave narrower protection and, in total, less money. We stop there rather than claiming a parallel result for value. The change in the value gradient does not clear conventional levels under wild-cluster inference, the per-mu gradient carries the weaker early estimate, and the early value cells support only ten of the twelve balanced provinces, so value is reported as a check on the interpretation of the architecture results rather than as a second headline (Appendix Table A5).

6.5 Type of land taken

The analysis sample retains every affirmative response to the expropriation question, so it includes homestead-only and other-usage cases alongside arable land. Two things would be required for this to generate the paper’s results: the mix would have to shift across cohorts, and it would have to be ordered by provincial development. Neither holds. The arable share moves from 95.1 percent in 2000–03 to 90.8 percent in the 2014+ cohort, a difference that does not clear conventional levels ($\chi^2 p = 0.075$), and across the fifteen provinces it is essentially uncorrelated with baseline fiscal pressure ($r = 0.065$). Re-estimating on the 2,041 arable-land households returns the same balanced twelve-province panel with twelve provinces in every cell. Compression is 36.4 percent for protection breadth, 32.9 for any durable protection, and 28.6 for cash only, against 37.8, 36.4, and 36.5 in the full sample, and every gradient change is larger and carries a smaller wild-cluster p -value than in Table 2 (Appendix Table A4). Cash-only compression is the one measure that moves materially; on this subsample it is better described as roughly a quarter to a third than as one-third.

6.6 Housing and the compensation-regime transition

Zhan (2026) dates a shift toward a housing-centered compensation regime to the late 2000s, the same period in which our provincial gradient erodes, so the compression reported above might be that shift in another guise. It is not, and the data answer directly on three counts. First, the primary outcome contains no housing at all: protection breadth counts pension, medical, and Dibao, and it shows the largest compression of the five measures at 37.8 percent. Second, housing is almost never decisive for the binary headline. Only 16 of 2,063 households, 0.8 percent, report housing as the only durable component received, and removing the housing item from any durable protection leaves compression at 31.8 percent against 36.4 percent with it included. Third, housing compensation does not in fact arrive late in this sample: receipt peaks at 7.6 percent among households expropriated in 2004–07 and falls to 1.8 percent among the 2014+ cohort. That last pattern is a limitation of our data rather than a refutation of Zhan’s account. Households resettled into urban housing are precisely those most likely to convert hukou and leave the rural sampling frame, so a rural survey is a poor instrument for observing a housing-centered regime. What the evidence supports is the narrower claim we need: within the rural frame, the compression is not generated by housing (Appendix Table A17).

6.7 Expropriation-specific content and the general expansion of rural social insurance

The period in which the provincial gradient erodes is also the period in which rural social insurance expanded nationally, so a natural objection is that what we observe is that expansion rather than anything specific to expropriation.

The outcomes are attributed by the respondent to the event. The questionnaire does not ask whether the household holds pension or medical insurance; it asks whether such insurance was transacted *due to this land expropriation*, and the same construction governs the cash, Dibao, housing, and employment items (Appendix Table A1). A household that enrolled in the New Rural Pension Scheme in the ordinary course does not, on this wording, count as having received pension protection in compensation. The measure is not a coverage rate, and a rise in general coverage does not mechanically move it.

A benchmark against rural households that were not expropriated is the obvious external check, but it is not commensurable with the comparison we run. Our unit is the province-cohort cell, where cohort is the timing of the household’s expropriation. Households never expropriated have no expropriation cohort, so the most that could be built is a province-by-survey-wave profile with two time points, to be set against a five-point cohort profile. The outcome variables differ as well: a general coverage item and an event-attributed compensation item are different measurements, not one measurement on two samples. We therefore report no such benchmark, and we do not think an informative one can be constructed from these data.

The wording that answers the objection creates a boundary we cannot test. If the propensity

to attribute an enrollment to the expropriation became more uniform across provinces as national schemes approached universality, then what converged is attribution rather than access. Nothing in CHIP separates the two, and we do not claim to. The claim we make is the one the measure supports: the protection households report receiving as a consequence of expropriation ceased to be ordered by provincial development.

6.8 Survey wave

The 2008–13 event cohorts are observed in both CHIP waves. Across the outcomes used in the audit, the development-gradient estimates have the same sign in both surveys and their confidence intervals overlap (Appendix Table A16). CHIP 2018 does not contain enough early-cohort support to replicate the full historical comparison on its own, but the overlapping cohorts show that the low-sorting later pattern is not specific to one survey wave.

6.9 Most recent expropriation and migration

The questionnaire asks multiply expropriated households to report the latest episode and does not contain a complete expropriation history. Earlier cohort membership therefore implicitly conditions on the absence of a subsequently reported expropriation. CHIP also records the survey province rather than a separate province of expropriation and does not provide the migration history needed to reconstruct event location. Finally, households that left the rural sampling frame before the survey are not observed. These restrictions mean that the estimates describe the geographic distribution among rural-frame households at the survey date and by the location observed in CHIP. We state these boundaries directly rather than treating them as empirically resolved.

6.10 Formal policy chronology

Province-wide framework dates add institutional context but do not provide independent identifying variation. Adoption timing is geographically structured, with coastal provinces adopting early or in the middle of the distribution and all late adopters in the CHIP sample located inland. Descriptively, correlations between formal timing and household-reported protection are modest and imprecise: across the pre-period, later period, and change comparisons, the absolute correlations do not exceed 0.42. A targeted audit also found no comparable province-wide operational date, distinct from the framework document date, that could be consistently constructed across the audited provinces. We therefore keep the chronology as descriptive appendix evidence and do not use it as an instrument or treatment timing variable. Appendix Table A14 reports the correlations.

6.11 Crisis exposure and competing province characteristics

The late-2000s boundary coincides with the global financial crisis and substantial regional differences in trade exposure. Pre-crisis trade openness is highly correlated with baseline fiscal pressure and GDP per capita across only fifteen provinces. Adding both dimensions attenuates the fiscal-gradient interaction, but the data cannot separately identify them (Appendix Table A13). This non-separability affects mechanism attribution, not the headline dispersion result, which is computed directly from realized province-level protection and does not use either provincial proxy.

7 Conclusion and Discussion

This paper has asked how the protection that replaces lost land security is distributed across provinces, and whether that distribution changed as China’s rural social-insurance system expanded. Using the harmonized land-expropriation modules of CHIP 2013 and 2018, we treat the architecture of compensation—a one-time cash payment, or a continuing claim on pension, medical, Dibao, housing, or employment protection—as the outcome to be explained rather than as a mechanism through which land loss later affects welfare.

Three findings follow. *First*, the early geography of protection was ordered by baseline provincial development, and steeply so. Across the interquartile range of a fixed 2000–02 measure of provincial fiscal pressure, households expropriated before 2008 in weaker-baseline provinces were about twenty percentage points less likely to receive any durable protection and about eighteen percentage points more likely to receive cash only. *Second*, that ordering then collapses. Among cohorts expropriated from 2008 onward the corresponding differences are economically close to zero and change sign, and baseline development explains 24 to 31 percent of cross-province variation in protection breadth among the early cohorts against 0 to 3 percent among the later ones. *Third*, total dispersion falls as well, by 36 to 38 percent in mean absolute pairwise provincial gaps on a balanced twelve-province panel, and this compression survives an explicit precision audit: equalizing cell sizes leaves a decline of 18 to 24 percent, while corrections that remove sampling variation imply a larger decline of 34 to 47 percent in latent between-province heterogeneity.

The second and third findings are economically distinct and it matters which is the larger. A fall in dispersion means only that provincial outcomes moved closer together. A collapse in the development gradient means that the particular hierarchy which organized early protection no longer predicts who receives it. Provinces in the later cohorts are not identical; their differences are simply no longer ranked by what those provinces looked like at the start of the period. In the vocabulary of the convergence literature, the β -side of this result is much the larger of the two, and it is the one we would ask a reader to take away.

We prefer a conservative interpretation of the timing. The evidence does not assign the

transition to a single national reform, and it cannot. Central requirements for land-expropriated farmers tightened over 2004–07, yet the development gradient remains steep through 2006–07; the subsequent period combines the expansion of rural pension and social-insurance infrastructure with the financial crisis and a large fiscal and credit response; and provincial trade exposure is strongly correlated with the development hierarchy across only fifteen provinces. Policy changes in this period occurred nationwide and do not provide an untreated counterfactual. We therefore read the evidence as institutional and descriptive rather than as a causal estimate of any mandate. What the data do support directly, in each cohort we observe, is that the geography of realized protection changed sharply around the late 2000s and that its systematic ordering by development did not survive the change.

Two implications for policy follow, neither of which requires attributing the change to a particular reform. The first concerns where the change came from. The gradient persisted through a period in which central requirements on local governments were tightening on paper, and eroded during a period in which nationwide social-insurance infrastructure was being built and integrated—the rural pension pilot from 2009, the Social Insurance Law in force from 2011, and the unification of the rural and urban resident pension schemes in 2014. Evidence from the parallel consolidation of urban and rural health insurance points the same way, in that integration changed what rural households actually used rather than only what was formally available to them (Huang and Wu 2020; Hao et al. 2025). A nationally co-financed scheme is in principle the kind of instrument that narrows a provincial capacity gradient in a way that an obligation addressed to local budgets does not; whether that is what operated here is beyond what this design can establish. The pattern is at least consistent with the view that building a common national infrastructure did more to equalize what land-losing households actually received than requirements addressed to local discretion did, and it suggests that obligations imposed without a shared delivery system are the weaker instrument. The second concerns what should be monitored. Because residual provincial variation is no longer ordered by development, indicators built on provincial economic capacity have lost much of their power to flag where households are underserved. Monitoring that asks whether a locality has established a program will therefore miss most of what remains; what is left to measure is whether individual households actually receive protection.

These findings broaden the distributional analysis of land expropriation. Existing studies examine downstream effects on income, saving, labor allocation, health, and political attitudes. Our results show that the institution replacing the lost security is itself unequally distributed before those downstream responses occur, so compensation architecture is part of the incidence of expropriation and not only a channel through which it operates. The same point applies more broadly to social policy: social protection is a distributional object before it becomes a redistributive instrument.

Three boundaries define what this evidence establishes, and none of them reaches the central

claim. We observe whether a household reports each form of protection, not what that protection is worth, so if benefit levels remained more dispersed across provinces than access did, our results *overstate* convergence in the welfare-relevant sense. What survives that caveat is the part that bears on incidence: the compensation amounts in Section 6 run in the same direction as the architecture among early cohorts, so weaker-baseline provinces were not buying narrower protection with larger payments. Households that left the rural sampling frame before the survey are not observed, and the outcome most likely to remove a land-losing household from it—hukou conversion with urban resettlement and entry into urban social insurance—is more available in stronger provinces. That exit depresses observed protection in strong provinces, so it *understates* the early development gradient: what we report is a floor on how steeply place once determined what a household received. The questionnaire records only the most recent expropriation, so early-cohort membership conditions on the absence of a later episode, and re-expropriation intensity is geographically structured; here the direction is genuinely open and we prefer to say so than to guess it. Administrative benefit records, a migration history, and a complete episode history would settle all three, and each is a well-defined target for work on data that contain them. None is required for what we report: within the rural frame CHIP observes, protection after land loss was once strongly ordered by provincial development, and among later cohorts it is not.

Appendix

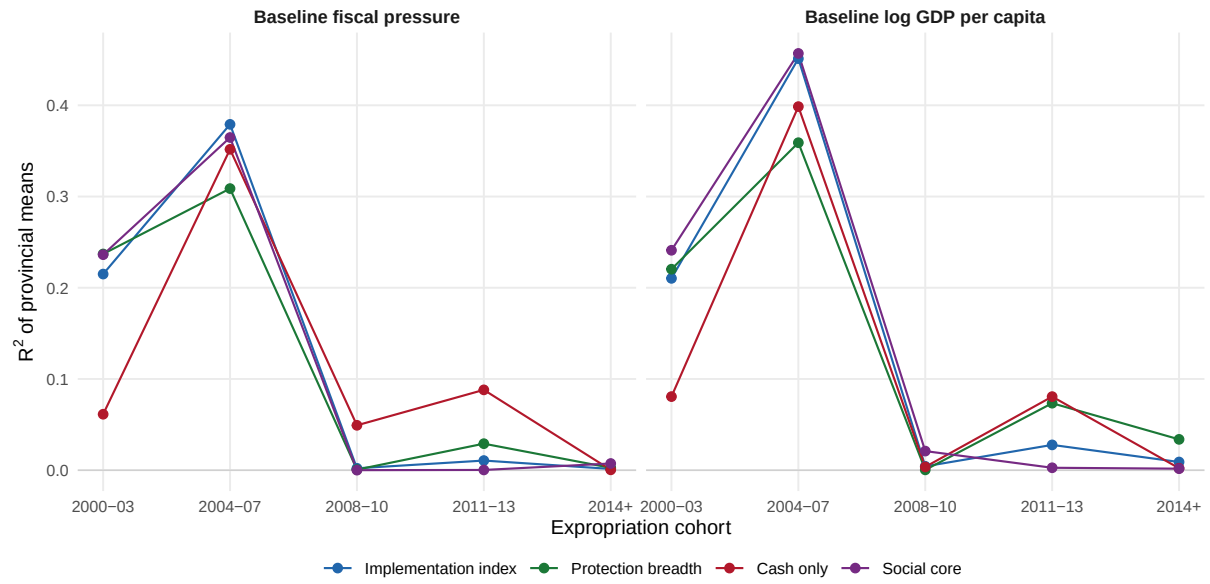


Figure A1: Share of cross-province variance in adjusted protection explained by one baseline measure at a time, balanced twelve-province panel.

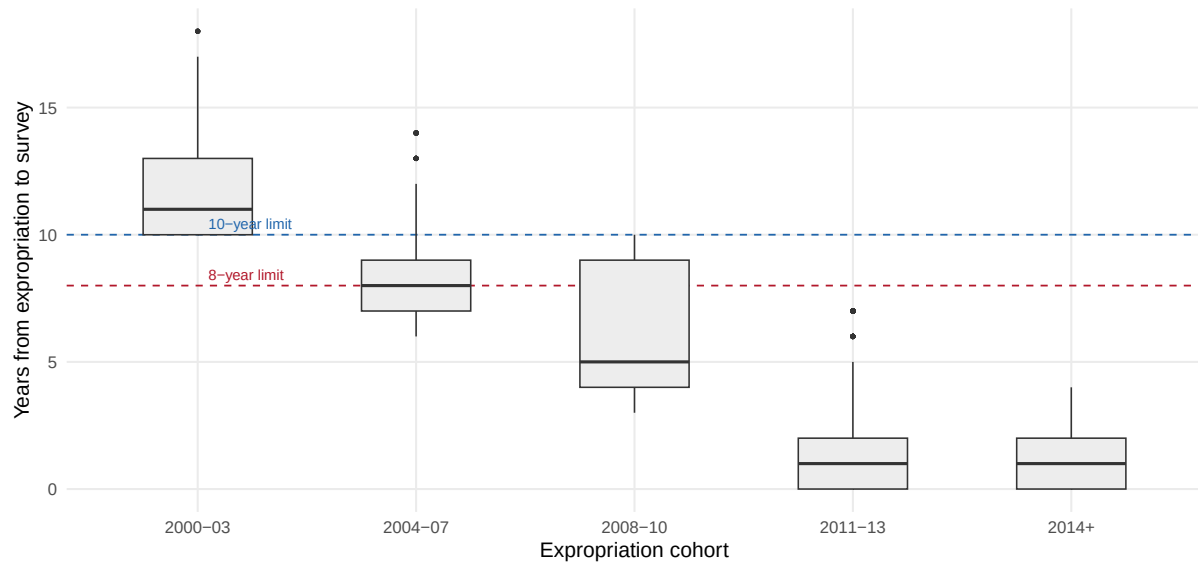


Figure A2: Event-to-survey lag by expropriation cohort, with the pre-specified lag restrictions marked.

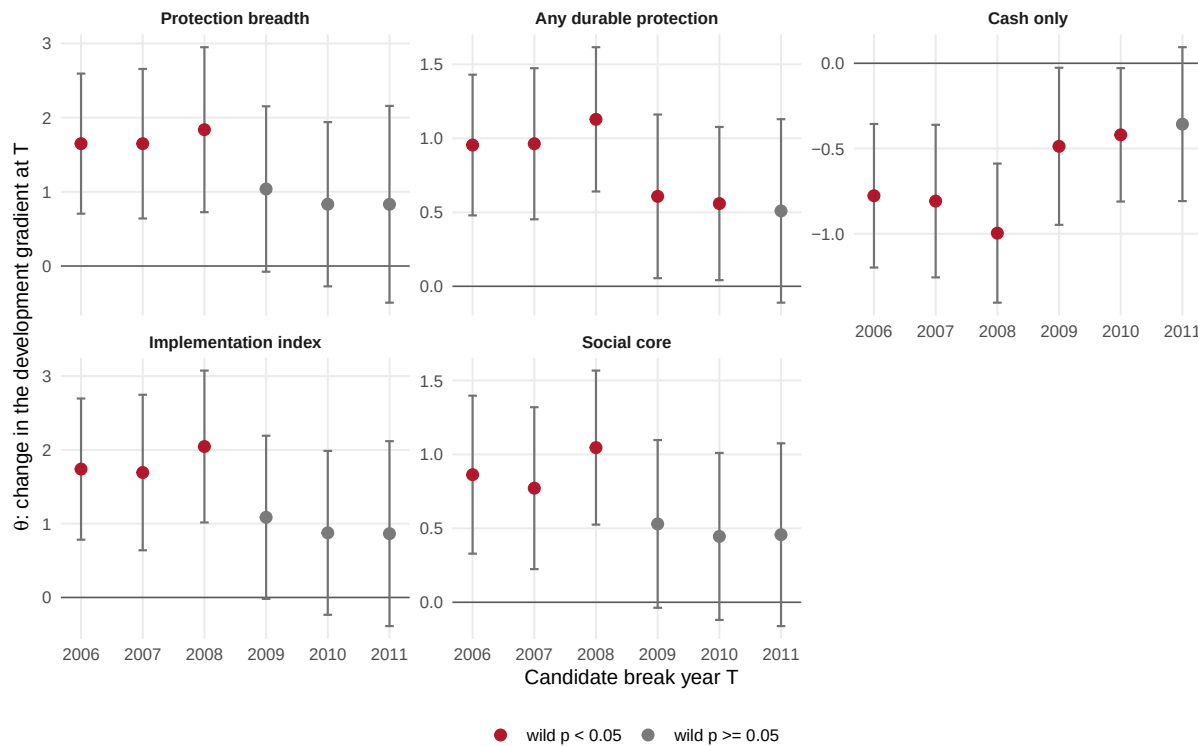


Figure A3: Change in the development gradient at each candidate break year, with province-clustered 95 percent confidence intervals. Points are shaded by whether the restricted wild-cluster bootstrap p -value falls below 0.05.

Table A1: Cross-wave questionnaire mapping for the land-expropriation module

CHIP 2013	CHIP 2018	Question (English questionnaire)	Harmonized variable	Enters
K05	K05	Did your household have the experience of land (collective-owned land) expropriation?	sample gate	sample definition
K06_1	K06_1	Date of the land expropriation (year, 4 digits). Households with more than one episode are instructed to report the latest one.	event year	cohort assignment
K06_2	K06_2	Size of the land expropriation (mu).	land area	descriptive only
K07_1	K07_1	Any compensation fees received (land compensation fee, resettlement subsidy, compensation for ground attachments and young crops) due to this land expropriation?	Cash	CashOnly, architecture
K07_2, K07_3	K07_2, K07_3	Total compensation obtained, and of which the amount the household itself received (yuan).	cash amount	descriptive only
K08_1	K08_1	Were there any pension insurance transacted for you or any member of your household due to this land expropriation?	Pension	Pension, SocialCore, ProtectionBreadth, AnyDurableProtection, ImplementationIndex
K08_2	K08_2	If any kind of pension insurance was transacted, what type is it?	pension type	descriptive only; not an outcome
K09_1	K09_1	Were there any medical insurance transacted for you or any household member due to this land expropriation?	Medical	Medical, SocialCore, ProtectionBreadth, AnyDurableProtection, ImplementationIndex
K09_2	K09_2	If any kind of medical insurance was transacted, what type is it?	medical type	descriptive only
not asked	K10	Have you had unemployment insurance due to this land expropriation?	not used	excluded by design
K10	K11	Did you or any household member apply for any minimum living guarantee (Dibao) due to this land expropriation?	Dibao	ProtectionBreadth, AnyDurableProtection, ImplementationIndex
K11	K12	Any housing arranged for you or any household member by the government due to this land expropriation?	Housing	AnyDurableProtection
K12	K13	Any replacement job arranged for you or any household member due to this land expropriation?	Employment	AnyDurableProtection

Notes. Question text is transcribed from the English rural questionnaires for CHIP 2013 and CHIP 2018. Neither wave's data file carries variable labels, so the questionnaires are the authority on wording. Item numbering shifts between waves because CHIP 2018 inserts an unemployment-insurance question at K10; that item has no CHIP 2013 counterpart and is excluded from every outcome so that the architecture is comparable across waves. The K08_2 pension-type option set also differs across waves and the harmonization maps the 2018 codes accordingly; no outcome in the paper uses pension *type*.

Table A2: Response options and coding rules for the harmonized items

Item (2013 / 2018)	Harmonized variable	Response options, CHIP 2013	Response options, CHIP 2018	Coding rule
K05	sample gate	1 Yes (both arable land and homestead); 2 Yes (arable land only); 3 Yes (homestead only); 4 Yes (land of other usage); 5 No	identical to 2013	sample gate: keep K05 in {1,2,3,4}; K05 = 5 leaves the module
K06.1	event year	free numeric year	free numeric year	event_year = K06.1; valid if 1949 <= year <= survey year; defines the cohort bins
K06.2	land area	free numeric	free numeric, one decimal	kept when strictly positive; otherwise missing
K07.1	Cash	1 Yes; 2 No	1 Yes; 2 No	Cash = 1 if K07.1 = 1, 0 if K07.1 = 2, missing otherwise
K07.2, K07.3	cash amount	free numeric	free numeric	K07.3 when non-negative, deflated to 2018 yuan by event-year CPI
K08.1	Pension	1 Yes; 2 No	1 Yes; 2 No	Pension = 1 if K08.1 = 1, 0 if K08.1 = 2, missing otherwise
K08.2	pension type	1 pension set up specifically for land-expropriated farmers; 2 urban pension (employee or resident); 3 new rural social insurance; 4 other	1 pension set up specifically for land-expropriated farmers; 2 resident endowment insurance; 3 new rural social insurance; 4 other	type 1 kept as land-lost-specific; 2013 code 2 maps to urban employee/resident, 2018 codes 2 and 3 both map to rural/resident because the 2018 option set merges them
K09.1	Medical	1 Yes; 2 No	1 Yes; 2 No	Medical = 1 if K09.1 = 1, 0 if K09.1 = 2, missing otherwise
K09.2	medical type	1 medical insurance specifically for land-expropriated farmers; 2 urban medical (employee or resident); 3 new rural cooperative medical; 4 other	1 medical insurance specifically for land-expropriated farmers; 2 resident; 3 new rural cooperative medical; 4 other	recorded but not used in any outcome
not asked / K10	not used	item does not exist in 2013	1 Yes; 2 No	asked only in 2018, so it is excluded from every outcome to keep the two waves comparable
K10 / K11	Dibao	1 Yes (urban Dibao); 2 Yes (rural Dibao); 3 No	identical to 2013	Dibao = 1 if the answer is 1 or 2, 0 if 3, missing otherwise
K11 / K12	Housing	1 Yes; 2 No	1 Yes; 2 No	Housing = 1 if the answer is 1, 0 if 2, missing otherwise
K12 / K13	Employment	1 Yes; 2 No	1 Yes; 2 No	Employment = 1 if the answer is 1, 0 if 2, missing otherwise

Notes. Options are transcribed from the English rural questionnaires; “identical to 2013” means the 2018 option set is unchanged. The rows are the items of Table A1, in the same order. Three harmonization decisions are visible here rather than only asserted: K05 is a sample gate rather than an outcome, so codes 1–4 all enter the analysis sample and code 5 leaves the module; Dibao is coded one from either the urban or the rural Dibao response; and the 2018 K08.2 option set merges two pension categories the 2013 set separates, which is why no outcome in the paper uses pension *type*. The 2018-only unemployment-insurance item at K10 is listed with its options so that the exclusion is visible rather than silent.

Table A3: Outcome definitions

Outcome	Definition	Scale	Role
Pension	K08_1 = yes	binary	component
Medical	K09_1 = yes	binary	component
Dibao	Dibao item = urban or rural Dibao	binary	component
Housing	housing item = yes	binary	component
Employment	replacement-job item = yes	binary	component
Cash	K07_1 = yes	binary	component
SocialCore	Pension OR Medical	binary	supporting evidence
CashSocialCore	Cash AND (Pension OR Medical)	binary	appendix
CashOnly	Cash AND NOT (Pension OR Medical OR Dibao)	binary	binary headline mirror
ProtectionBreadth	Pension + Medical + Dibao (0-3 count)	count 0-3	primary outcome
AnyDurableProtection	1 if any of Pension, Medical, Dibao, Housing, Employment	binary	binary headline
ImplementationIndex	equally weighted z-score average of SocialCore, CashSocialCore, Pension, Medical, ProtectionBreadth and reversed CashOnly	standardized	summary robustness measure

Notes. Each outcome is a formula over the harmonized items in Table A1. No household coded as cash only reports pension, medical, Dibao, employment, or housing protection.

Table A4: Type of land expropriated and the arable-land subsample

<i>Panel A. Composition by cohort (percent of households)</i>					
Cohort	<i>N</i>	Arable and homestead	Arable only	Homestead only	Other usage
2000-03	185	10.8	84.3	0.5	4.3
2004-07	330	19.4	76.4	1.2	3.0
2008-10	432	15.5	77.5	0.9	6.0
2011-13	572	15.2	78.7	1.0	5.1
2014+	544	13.1	77.8	1.5	7.7
Full sample	2180	15.4	78.2	1.1	5.3

<i>Panel B. Headline objects re-estimated on arable-land households</i>					
Outcome	MAPG compression (%)		Gradient change $\hat{\theta}$		
	Full	Arable	Full	Arable	Wild <i>p</i> , arable
Protection breadth	37.8	36.4	1.647	1.945	0.0040
Any durable protection	36.4	32.9	0.933	1.125	0.0090
Cash only	36.5	28.6	-0.822	-1.001	0.0125
Implementation index	37.0	35.7	1.473	1.752	0.0050
Social core	30.2	31.8	0.654	0.782	0.0775

Notes. The expropriation question records whether arable land, the homestead, or land of other usage was taken; the analysis sample retains all four affirmative responses. Panel A gives the composition of the 2,180-household architecture sample; the row labeled full sample pools every cohort, including the 117 households expropriated before 2000. Panel B compares the full sample with the 2,041 households whose expropriation involved arable land, that is categories one and two of Panel A. Compression is the percentage decline in the mean absolute pairwise provincial gap from the two earlier cohorts to the three later ones on the balanced twelve-province panel, which the arable subsample reproduces with twelve provinces in every province-cohort cell. $\hat{\theta}$ is the coefficient on $D_p \times Post_t$ in equation (4). Full-sample columns are those of Tables 3 and 2; the arable columns are estimated here with the same specification and the same restricted wild-cluster bootstrap, 1,999 draws over fifteen province clusters. Across the fifteen provinces the arable share is essentially uncorrelated with baseline fiscal pressure ($r = 0.065$), and its variation across cohorts does not clear conventional levels ($\chi^2 p = 0.075$).

Table A5: Compensation value and the baseline development gradient

Panel A. Reported amounts by cohort (2018 yuan)						
Cohort	<i>N</i>	Reporting an amount (%)	Median real compensation		Median per mu	
2000-03	185	84.9	15,013		11,913	
2004-07	330	90.3	20,720		15,731	
2008-10	432	94.2	23,656		13,696	
2011-13	572	90.9	27,489		19,626	
2014+	544	87.9	15,156		14,518	
Panel B. Value gradient on baseline fiscal pressure						
Outcome	Expropriated before 2008		Expropriated 2008 onward		Pooled change	
	$\hat{\beta}$ (SE)	IQR	$\hat{\beta}$ (SE)	IQR	$\hat{\theta}$	Wild <i>p</i>
Log real compensation	-1.782 (0.422)	-0.369	1.484 (0.902)	+0.307	3.452	0.2345
Log real compensation per mu	-0.524 (0.613)	-0.108	2.167 (1.177)	+0.449	2.199	0.5335

Notes. Compensation is the amount the household reports receiving, deflated to 2018 yuan using event-year CPI; per-mu divides it by the reported area taken. Both are conditional on a positive reported amount. Panel B estimates equation (3) separately on the two earlier cohorts and the three later ones, with expropriation-year and survey-wave effects and province-clustered standard errors. “IQR” converts the coefficient into the difference associated with moving from the 25th to the 75th percentile of baseline fiscal pressure across the fifteen provinces, in log points; a negative entry means weaker-baseline provinces paid less. $\hat{\theta}$ is the pooled change in the gradient from equation (4), with a restricted wild-cluster bootstrap *p*-value over fifteen province clusters, 1,999 draws. Neither pooled change clears conventional levels, so value is reported as a check on the interpretation of the architecture results rather than as a second headline.

Table A6: Cohort-specific geographic dispersion, balanced twelve-province panel

Outcome	Mean absolute pairwise provincial gap					Compression (%)		Full sample
	2000–03	2004–07	2008–10	2011–13	2014+	MAPG	SD	MAPG (%)
Protection breadth	0.522	0.478	0.310	0.343	0.280	37.8	41.3	27.1
Any durable protection	0.283	0.314	0.179	0.189	0.202	36.4	35.9	25.3
Cash only	0.295	0.265	0.189	0.184	0.160	36.5	34.5	24.6
Implementation index	0.531	0.542	0.333	0.346	0.336	37.0	40.7	25.0
Social core	0.256	0.316	0.199	0.191	0.208	30.2	32.1	18.5

Notes. Mean absolute pairwise provincial gaps among adjusted province-cohort means. The main text summarizes these cohort paths and reports precision-adjusted compression measures.

Table A7: Cohort-specific development gradients

Cohort	Gradient	(SE)	95% CI	<i>p</i>	<i>N</i>
<i>Protection breadth</i>					
2000–03	-2.029***	(0.737)	[-3.47, -0.59]	0.006	185
2004–07	-1.555***	(0.548)	[-2.63, -0.48]	0.005	330
2008–10	0.334	(0.326)	[-0.30, 0.97]	0.305	432
2011–13	0.603	(0.658)	[-0.69, 1.89]	0.359	572
2014+	0.054	(0.263)	[-0.46, 0.57]	0.838	544
<i>Any durable protection</i>					
2000–03	-0.996***	(0.345)	[-1.67, -0.32]	0.004	185
2004–07	-0.952***	(0.271)	[-1.48, -0.42]	<0.001	330
2008–10	0.235	(0.189)	[-0.14, 0.60]	0.214	432
2011–13	0.398	(0.356)	[-0.30, 1.10]	0.264	572
2014+	-0.024	(0.281)	[-0.57, 0.53]	0.932	544
<i>Cash only</i>					
2000–03	0.827**	(0.330)	[0.18, 1.47]	0.012	185
2004–07	0.903***	(0.245)	[0.42, 1.38]	<0.001	330
2008–10	-0.277	(0.194)	[-0.66, 0.10]	0.154	432
2011–13	-0.445	(0.353)	[-1.14, 0.25]	0.208	572
2014+	0.216	(0.228)	[-0.23, 0.66]	0.344	544
<i>Implementation index</i>					
2000–03	-2.083***	(0.746)	[-3.54, -0.62]	0.005	185
2004–07	-1.934***	(0.518)	[-2.95, -0.92]	<0.001	330
2008–10	0.281	(0.332)	[-0.37, 0.93]	0.398	432
2011–13	0.555	(0.691)	[-0.80, 1.91]	0.422	572
2014+	-0.206	(0.374)	[-0.94, 0.53]	0.581	544
<i>Social core</i>					
2000–03	-1.068***	(0.403)	[-1.86, -0.28]	0.008	185
2004–07	-1.059***	(0.242)	[-1.53, -0.58]	<0.001	330
2008–10	0.063	(0.178)	[-0.29, 0.41]	0.725	432
2011–13	0.229	(0.351)	[-0.46, 0.92]	0.514	572
2014+	-0.191	(0.263)	[-0.71, 0.33]	0.469	544

Notes. Within-cohort gradients on fixed 2000–02 provincial fiscal pressure with expropriation-year and survey-wave effects where applicable. Province-clustered standard errors. The 2000–02 baseline overlaps the 2000–03 cohort; the 2004–07 cohort provides the clean early benchmark for temporal-precedence statements.

Table A8: Candidate-breakpoint profile for the change in the development gradient

Outcome	2006	2007	2008	2009	2010	2011
Protection breadth	1.65 [0.005]	1.65 [0.007]	1.84 [0.001]	1.04 [0.105]	0.83 [0.207]	0.83 [0.302]
Any durable protection	0.95 [0.019]	0.96 [0.010]	1.13 [0.001]	0.61 [0.045]	0.56 [0.030]	0.51 [0.198]
Cash only	-0.78 [0.015]	-0.81 [0.011]	-1.00 [0.002]	-0.49 [0.048]	-0.42 [0.040]	-0.36 [0.176]
Implementation index	1.74 [0.004]	1.69 [0.013]	2.05 [0.001]	1.09 [0.099]	0.88 [0.172]	0.87 [0.240]
Social core	0.86 [0.008]	0.77 [0.032]	1.05 [0.002]	0.53 [0.089]	0.45 [0.160]	0.46 [0.216]

Notes. Each entry is $\hat{\theta}$ from a single-break specification $Y = \text{FiscalPressure}_p \times \mathbf{1}[t \geq T]$ with province, expropriation-year, and survey-wave effects, estimated on households with expropriation years from 2000 onward. Restricted wild-cluster bootstrap p -values in brackets, 999 draws, fifteen province clusters. The single-break reference group pools all cohorts before T and therefore differs from the three-period specification in Table 2.

Table A9: Fine timing of the development gradient

Outcome	2000–03	2004–05	2006–07	2008–10	2011–13	2014+
Implementation index	-2.083	-1.885	-1.884	0.281	0.555	-0.206
Protection breadth	-2.029	-1.402	-1.591	0.334	0.603	0.054
Cash only	0.827	0.868	0.904	-0.277	-0.445	0.216
Social core	-1.068	-1.096	-0.981	0.063	0.229	-0.191

Notes. Within-cohort gradient levels, not changes, on fixed 2000–02 fiscal pressure, splitting 2004–07 into 2004–05 and 2006–07. Support is 151 and 179 households across fourteen and fifteen provinces. The 2006–07 estimate has the smallest standard error of the three pre-2008 bins.

Table A10: Retrospective-lag restrictions, common 2004–07 early window

<i>Panel A. MAPG compression</i>			
Outcome	Unrestricted	≤ 10 years	≤ 8 years
Implementation index	37.6	42.0	44.7
Protection breadth	34.9	39.1	42.1
Cash only	32.9	37.4	44.3
Social core	36.8	40.5	38.3
<i>Panel B. Early development gradient under the same restrictions</i>			
Outcome	Unrestricted $\hat{\beta}$ (SE)	≤ 10 years $\hat{\beta}$ (SE)	≤ 8 years $\hat{\beta}$ (SE)
Protection breadth	-1.555 (0.548)	-2.090 (0.629)	-1.747 (0.673)
Any durable protection	-0.952 (0.271)	-1.346 (0.310)	-1.138 (0.307)
Cash only	0.903 (0.245)	1.196 (0.281)	1.104 (0.301)
Implementation index	-1.934 (0.518)	-2.592 (0.591)	-2.243 (0.615)
Social core	-1.059 (0.242)	-1.443 (0.270)	-1.254 (0.269)

Notes. MAPG compression in percent, balanced twelve-province panel. The early window is held at 2004–07 in every column, because the 2000–03 cohort cannot survive any lag restriction. A five-year limit removes the entire pre-2008 era and is therefore not reported. Panel B estimates equation (3) on the 2004–07 cohort alone under the same restrictions, with expropriation-year effects, a survey-wave effect where both waves survive the restriction, and province-clustered standard errors. Support falls from 330 households in 15 provinces across both waves to 257 in 14 provinces and 180 in 14 provinces; both restricted columns retain only the CHIP 2013 wave, because a 2004–07 expropriation observed in 2018 is at least eleven years old. The restricted estimates are therefore not more informative than the unrestricted one, only differently selected.

Table A11: Compression after reweighting later cohorts to early-cohort composition

Outcome	MAPG compression (%)		SD compression (%)	
	Baseline	Rewighted	Baseline	Rewighted
Implementation index	37.0	32.3	40.7	36.1
Protection breadth	37.8	32.9	41.3	37.3
Cash only	36.5	27.8	34.5	25.3
Social core	30.2	26.5	32.1	27.2

Notes. Later cohorts reweighted by inverse-probability weights from a logit on sex, education, age at expropriation, and its square. Standardized mean differences fall from -0.65 to $+0.02$ on age at expropriation; the effective sample is 877 of 1,547 later households and the largest weight is 3.74. No trimming.

Table A12: Coarse and fine compensation measures

Class	Outcome	MAPG before	MAPG after	Compression (%)
Coarse	Cash only	0.280	0.178	36.5
Coarse	Any durable protection	0.299	0.190	36.4
Fine	Medical	0.281	0.166	40.7
Fine	Cash + social core	0.260	0.161	38.3
Fine	Protection breadth	0.500	0.311	37.8
Fine	Implementation index	0.537	0.338	37.0
Fine	Social core	0.286	0.200	30.2
Fine	Pension	0.245	0.189	22.7

Notes. Balanced twelve-province panel. Coarse measures require only the distinction between a cash-only package and one containing any durable component. “Any durable protection” is an indicator for at least one of pension, medical, Dibao, employment, or housing.

Table A13: Pre-crisis trade exposure and the post-2008 gradient change

Outcome	Gradient change, 2008 onward		Magnitude reduced (%)	Wild-cluster p	
	Baseline	With exposure		Baseline	With exposure
Implementation index	1.891	0.989	48	0.004	0.169
Protection breadth	1.983	1.462	26	0.005	0.049
Cash only	-0.757	-0.287	62	0.049	0.396
Social core	0.895	0.241	73	0.033	0.462

Notes. Pre-crisis trade openness is provincial goods imports plus exports over GDP, averaged over 2005–07. It correlates with baseline fiscal pressure at -0.866 and with baseline log GDP per capita at $+0.880$ across the fifteen provinces. Wild-cluster p -values from 1,999 restricted bootstrap draws. The baseline column is estimated on the 2,063 households with expropriation years from 2000 onward, so that the comparison with and without trade exposure is made on a common sample. It therefore differs from $\hat{\theta}$ in Table 2, which is estimated on all 2,180 households and whose pre-2004 reference group also contains the 117 households expropriated before 2000: 1.983 against 1.647 for protection breadth, -0.757 against -0.822 for cash only, and 1.891 against 1.473 for the implementation index. The sign, the ordering across outcomes, and the conclusion are unaffected by which of the two samples is used.

Table A14: Formal provincial framework timing and realized household protection

Outcome	Before 2008		2008 onward		Change	
	r	p	r	p	r	p
Implementation index	-0.286	0.321	0.229	0.431	0.374	0.188
Protection breadth	-0.211	0.469	0.285	0.323	0.323	0.259
Cash only	0.035	0.906	-0.266	0.357	-0.202	0.488
Social core	-0.375	0.187	0.134	0.648	0.418	0.137
Pension	-0.147	0.617	0.062	0.834	0.176	0.548

Notes. Pearson correlations between provincial formal framework year and demographically adjusted provincial means over fourteen provinces with usable cells on both sides of 2008. Formal timing is used descriptively and not as an instrument or treatment date.

Table A15: Inference and province sensitivity for geographic compression

<i>Panel A. Bootstrap inference for MAPG compression</i>					
Outcome	Raw MAPG compression (%)	Household-resampling 95% CI		Province-resampling 95% CI	
Protection breadth	37.8	[19.4, 46.8]		[-22.9, 57.0]	
Any durable protection	36.4	[20.2, 46.5]		[4.2, 52.1]	
Cash only	36.5	[20.9, 46.3]		[6.1, 53.7]	
Implementation index	37.0	[20.3, 46.6]		[-14.8, 55.6]	
Social core	30.2	[13.9, 41.8]		[-16.5, 50.5]	
<i>Panel B. Leave-one-province-out compression</i>					
Outcome	Full (%)	Min (%)	Omitted	Max (%)	Omitted
Protection breadth	37.8	19.9	Jiangsu	42.8	Hunan
Any durable protection	36.4	27.8	Jiangsu	40.7	Chongqing
Cash only	36.5	27.3	Jiangsu	44.5	Chongqing
Implementation index	37.0	21.2	Jiangsu	42.4	Anhui
Social core	30.2	14.7	Jiangsu	37.2	Anhui

Notes. Balanced twelve-province panel. Compression is the percentage decline in the mean absolute pairwise provincial gap from the average of the two earlier cohorts to the average of the three later ones; the raw column reproduces Table 3. In Panel A, household resampling draws households with replacement within each province-cohort cell and holds the twelve provinces fixed; province resampling draws the twelve provinces with replacement, treating them as draws from a broader population. Both use 2,000 draws, percentile intervals at the 2.5th and 97.5th quantiles, and seed 20260825. Demographic residualization is estimated once on the original sample and held fixed across draws rather than recomputed within each draw. Panel B omits one of the twelve provinces at a time and recomputes compression on the remaining eleven, retaining the same residualization, five-household cell floor, cohort grouping, equal province weighting, and MAPG definition; it is a sensitivity diagnostic and not an inferential procedure.

Table A16: Development gradients in overlapping cohorts, by survey wave

<i>Panel A. 2008–10 cohort</i>				
Outcome	CHIP 2013 $\hat{\beta}$ (SE)	95% CI	CHIP 2018 $\hat{\beta}$ (SE)	95% CI
Protection breadth	0.314 (0.532)	[-0.729, 1.356]	0.365 (0.374)	[-0.368, 1.098]
Any durable protection	0.192 (0.270)	[-0.336, 0.721]	0.310 (0.288)	[-0.254, 0.875]
Cash only	-0.262 (0.273)	[-0.796, 0.273]	-0.302 (0.284)	[-0.858, 0.255]
Implementation index	0.252 (0.526)	[-0.780, 1.283]	0.318 (0.506)	[-0.673, 1.309]
Social core	0.039 (0.257)	[-0.466, 0.543]	0.097 (0.299)	[-0.489, 0.684]
<i>Panel B. 2011–13 cohort</i>				
Outcome	CHIP 2013 $\hat{\beta}$ (SE)	95% CI	CHIP 2018 $\hat{\beta}$ (SE)	95% CI
Protection breadth	0.670 (0.655)	[-0.614, 1.954]	0.547 (0.949)	[-1.313, 2.407]
Any durable protection	0.397 (0.360)	[-0.309, 1.103]	0.549 (0.570)	[-0.568, 1.665]
Cash only	-0.454 (0.364)	[-1.168, 0.261]	-0.549 (0.570)	[-1.665, 0.568]
Implementation index	0.614 (0.689)	[-0.736, 1.965]	0.580 (1.034)	[-1.446, 2.606]
Social core	0.252 (0.355)	[-0.444, 0.947]	0.271 (0.501)	[-0.712, 1.254]

Notes. The two expropriation cohorts observed in both CHIP waves. Each estimate is the wave-specific analogue of equation (3): the outcome on fixed 2000–02 provincial fiscal pressure with expropriation-year effects and province-clustered standard errors, estimated within a single survey wave so that no survey-wave effect enters. Confidence intervals are the conventional 95 percent normal intervals on the clustered standard error. Support is 261 households in CHIP 2013 and 171 in CHIP 2018 for the 2008–10 cohort, and 448 and 124 for 2011–13; fourteen provinces in every cell. The sign agrees across waves and the intervals overlap in all ten outcome-cohort comparisons.

Table A17: Housing compensation and the robustness of durable-protection compression

<i>Panel A. Housing receipt by expropriation cohort</i>		
Cohort	<i>N</i>	Housing receipt (%)
2000-03	185	2.7
2004-07	330	7.6
2008-10	432	3.7
2011-13	572	4.7
2014+	544	1.8
<i>Panel B. Sensitivity of the binary durable measure to housing</i>		
Measure	MAPG compression (%)	
Any durable protection, baseline definition	36.4	
Any durable protection, excluding housing	31.8	
Protection breadth (never includes housing)	37.8	

Notes. Panel A covers the 2,063 households in the five cohort bins. Panel B recomputes MAPG compression on the balanced twelve-province panel with the housing item removed from any durable protection, retaining pension, medical, Dibao, and employment; protection breadth is unchanged because it never includes housing. Sixteen of 2,063 households, 0.8 percent, report housing as the only durable component they received.

Data availability

Data are available upon reasonable request, subject to applicable CHIP data-use restrictions.

Declaration of generative AI and AI-assisted technologies in the writing process

ChatGPT (OpenAI) and Claude (Anthropic) were used to assist with manuscript preparation and language editing. The author reviewed and edited all AI-assisted content and takes full responsibility for the article.

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